

HANDOUTS

MGT201

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Detailed explanation of Ppt Slides provided by VU , made with the help of AI.

NOTE = If you go through the VU PPT slides of mgt201 once before studying this handout, it will help you understand the topics more effectively and develop clearer concepts.

Module 1: Scope of Financial Management:

I. Functions of Financial Management

The functions of financial management represent the primary pillars upon which a firm's value is built. These functions are categorized into four main areas: Financing, Investing, Asset Management, and Financial Analysis.

1. Financing Activities

Financing activities focus on how a company raises the capital necessary to fund its operations and growth. This involves deciding between **Internal Financing** (retained earnings/profits) and **External Financing** (funds from outside investors or lenders).

- **Equity vs. Debt:** A critical decision is the mix of equity (ownership) and debt (borrowed funds). **Equity Financing** involves issuing common or preferred shares, which grants certain rights and dividends to shareholders but does not require mandatory repayment. **Debt Financing** involves public issues or private loans that carry interest obligations.
- **Financial Leverage:** This refers to the use of debt to potentially increase the return on equity. While it can magnify profits, it also increases the risk of the firm due to fixed interest obligations.

2. Investing Activities

Investing involves the strategic allocation of the capital raised through financing into assets that will generate future returns.

- **Terms of Investment:** Investments can be **Long-Term** (fixed assets, infrastructure) or **Short-Term** (marketable securities).
- **Capital Budgeting:** This is the process of evaluating long-term investment projects using **Discounted Cash Flow (DCF)** methods to determine their viability based on the required **Rate of Return**.
- **External Markets:** Companies can also invest outside their core business in **Stock Markets** (like the PSX) or through **Indirect Investment** such as **Mutual Funds**.

3. Asset Management (Working Capital Management)

Asset management ensures that the firm's daily operations run smoothly by managing its short-term resources.

- **Components:** The primary components are **Inventories**, **Receivables**, and **Cash**.
- **Goal:** The objective is to maintain an optimal level of **Net Working Capital** and monitor performance through **Working Capital Ratios** (Current and Acid Test Ratios) and the **Cash/Credit Cycle**.

4. Financial Analysis

Financial analysis is the process of evaluating a company's performance to support decision-making.

- **Types of Analysis:** This includes **Comparative Analysis** (comparing performance over time), **Index Analysis**, and **Risk Analysis**.
- **Ratio Analysis:** This is a vital tool used to assess **Liquidity** (ability to meet short-term debts), **Leverage** (debt levels), **Turnover** (efficiency of asset use), and **Profitability**.

II. Detailed Breakdown of Financing Activities

1. Sources and Types of Financing

Financing is generally categorized by its source and time period. **Long-term financing** typically involves equity or long-term debt used for major assets, while short-term financing handles operational needs.

- **Equity Financing (Common vs. Preferred):** Common shares represent ownership and usually come with voting rights. Preferred shares often have a fixed **dividend** rate and priority over common shares during liquidation.
- **Internal Financing:** This is the use of the company's own **Profit** to reinvest in the business, which avoids the cost of issuing new securities.

2. Debt Financing and Interest Rates

Debt involves borrowing funds that must be repaid with interest.

- **Interest Structures:** Loans can have **Fixed Rates** (staying the same) or **Floating Rates** (changing based on benchmarks like **KIBOR** or the Prime Rate).
- **(KIBOR)=KARACHI INTER BANK OFFERED RATE.**
- **Public vs. Private:** Companies can raise debt through **Public Issues** (bonds) or **Private Loans** from financial institutions.

III. Detailed Breakdown of Asset Management

1. Inventory Management

Inventory represents a significant portion of current assets. It is categorized into **Raw Materials**, **Work in Process (WIP)**, and **Finished Goods**, alongside stores and spares.

- **Inventory Costs:** Managers must balance **Ordering Costs** (the cost of placing an order) and **Carrying Costs** (the cost of holding stock in the warehouse) to find the most efficient stock level.

3. Receivables Management

Receivables are the amounts owed to a firm by its customers.

- **Credit Policy:** A firm must define its **Credit Period** (how long customers have to pay) and **Cash Discounts** (incentives for early payment).
- **Risks:** Managers must account for **Bad Debts** (uncollectible accounts) and the **Collection Cost** associated with recovering money.
- **Factoring:** This is a specialized strategy where a firm sells its receivables to a third party (a factor) to get immediate cash.

4. Cash Management

Cash is the most liquid asset and is essential for solvency.

- **Cash Budget:** This tool tracks **Cash Inflows and Outflows** to predict **Surplus or Deficits**.
- **Technical Insolvency:** This occurs when a firm cannot meet its immediate financial obligations despite having assets, highlighting the importance of liquidity over simple profitability.

IV. Detailed Breakdown of Financial Analysis

1. Ratio Analysis Categories

To truly understand a firm's health, financial managers look at several ratio categories:

- **Liquidity Analysis:** Measures the ability to pay short-term debts.
- **Turnover/Activity Analysis:** Measures how efficiently a firm uses its assets to generate sales.
- **Leverage/Gearing Analysis:** Examines the proportion of debt in the capital structure.
- **Profitability and Market Analysis:** Assesses the firm's ability to generate earnings relative to sales or share price.

Top 5 Important Questions

1. **Explain the trade-off between risk and return in the context of Financial Leverage. How does the use of debt financing impact the firm's profitability and its risk of technical insolvency?**

Financial leverage refers to the use of debt financing (such as private loans or public issues) to fund a firm's operations.

Financial leverage means the use of **debt financing** to increase shareholders' return. The main point is that debt can increase profitability when the firm earns more than the cost of interest. In this case, shareholders enjoy a higher return on equity. However, debt also increases risk because interest payment is compulsory whether the firm earns profit or not. If business income falls, the company may face losses and cash shortage. Therefore, the trade-off is that **higher debt increases both return and risk**. Too much debt can also cause **technical insolvency**, where the firm cannot meet short-term obligations on time.

2. **Compare and contrast Equity and Debt as sources of long-term financing. What are the implications for a company's control (voting rights) and its mandatory financial obligations (interest vs. dividends)?**

2. Equity vs Debt Financing

Equity and debt are important long-term sources of finance. **Equity financing** means issuing common or preferred shares, which gives investors ownership rights and, in the case of common shares, voting rights as well. Dividends are paid from profit and are not compulsory. On the other hand, **debt financing** means borrowing funds through loans or bonds. Lenders do not get ownership or voting rights, but the company must pay fixed interest and repay the principal amount on maturity. The main point is that equity reduces financial risk but may dilute control, while debt keeps ownership control safe but increases financial obligations and risk.

3. **Discuss the components of the Credit Policy in Receivables Management. How does a firm balance the benefits of offering cash discounts and longer credit periods against the risks of bad debts and increased collection costs?**

3) Credit Policy in Receivables Management

Credit policy is the set of rules that a firm uses when selling goods on credit. Its main points include **credit period, cash discounts, and credit standards**. A longer credit period and attractive cash discounts can increase sales because customers get more convenience. However, this also increases the chances of **bad debts, delayed payments, and collection costs**. Therefore, the financial manager must maintain a balance between the benefit of increased sales and the cost of non-payment risk. A good credit policy improves receivables management and helps maintain smooth cash flow.

4. **Analyze the importance of the Cash Budget in preventing technical insolvency. How does a manager use the forecast of inflows and outflows to make decisions about short-term financing and lending?**

4) Importance of Cash Budget

A cash budget is an important tool that estimates future **cash inflows and cash outflows**. The main purpose is to help the manager know whether the firm will have a **cash surplus or deficit** in the coming period. On this basis, the company can plan short-term borrowing when cash is low or short-term investment when extra cash is available. This helps in maintaining liquidity and meeting daily expenses such as salaries, suppliers, and utility payments. The key point is that a proper cash budget prevents **technical insolvency** and keeps the business financially stable.

5. Evaluate the role of Capital Budgeting and Discounted Cash Flow (DCF) methods in Investing Activities. Why is the "Rate of Return" a critical factor when deciding between internal projects and external market investments?

5) Capital Budgeting and DCF Methods

Capital budgeting is the process of evaluating long-term investment projects such as machinery, buildings, or expansion plans. The main point is that firms use **Discounted Cash Flow (DCF) methods** to calculate the present value of future cash inflows and compare it with the initial investment. The **rate of return** is critical because it shows whether the project will generate enough profit compared to its cost and risk. It also helps managers compare internal projects with external investments like stock market opportunities. Therefore, capital budgeting ensures that company funds are invested in the most profitable projects.

MODULE 2 (13 TO 22) : Introduction to Financial Environment:

Topic 013 & 014: The Financial Environment and the Central Bank (SBP)

The **financial environment** constitutes the external framework within which businesses operate, encompassing regulators, financial markets, and credit providers. For a management student, understanding this environment is crucial because it dictates the availability of capital and the regulatory constraints on business operations. The environment is categorized into three primary pillars: **Regulators** (such as the SECP and SBP) who ensure systemic stability; **Credit Providers** (Banks and NBFIs) who facilitate liquidity; and **Financial Markets** where securities are traded.

At the heart of this environment is the **Central Bank**, specifically the **State Bank of Pakistan (SBP)**. The SBP serves several critical functions that maintain economic equilibrium. Firstly, it is the **Bank of Issue**, possessing the sole authority to print and manage the national currency. Beyond currency management, it acts as the **Banker, Agent, and Advisor to the Government**, managing public debt and advising on economic policy. In a regulatory capacity, the SBP is the **Custodian of Cash Reserves** for commercial banks and the **Custodian of Foreign Balances**, ensuring the country has enough foreign exchange to facilitate international trade.

One of the SBP's most vital roles in management theory is acting as the **Lender of Last Resort**. This means that in times of financial distress or liquidity crises, the SBP provides funds to commercial banks to prevent a total collapse of the banking system. Furthermore, it functions as a **Clearing House** for inter-bank transactions and serves as the **Controller of Credit**, using monetary policy tools (like interest rates) to manage inflation and economic growth. Ultimately, its primary goal is the **protection of depositors' interests**, ensuring public trust in the financial system remains intact.

Key Takeaways for Revision:

- **Financial Environment Components:** Regulators, Credit Providers, and Markets.
- **SBP Primary Functions:** Currency issuance, government advisor, and custodian of reserves.
- **Systemic Stability:** Acts as the "Lender of Last Resort" to prevent bank runs and financial collapse.
- **Credit Control:** Manages the volume and cost of credit in the economy to control inflation.

Topic 015 & 016: SECP and the Federal Board of Revenue (FBR)

The **Securities and Exchange Commission of Pakistan (SECP)** is the primary regulator for the **corporate sector and capital markets**. While the SBP focuses on the banking sector, the SECP ensures that companies, insurance providers, and stock markets operate transparently. Its core mission is to **support investment, economic growth, and prosperity** by creating a fair and efficient regulatory framework.

The SECP's functions are multifaceted: it handles the **registration and licensing** of companies, performs **prudential supervision** to ensure businesses are financially sound, and operates an **Appellate Bench** to resolve legal disputes. Notably, it also manages **investor awareness and education**, which is essential for mobilizing domestic savings into productive corporate investments.

Complementing these regulatory efforts is the **Federal Board of Revenue (FBR)**, which is the cornerstone of Pakistan's **Fiscal Policy**. While the SBP manages money (monetary policy), the FBR manages government revenue through **taxation**. The FBR's primary mandate is to **levy and collect taxes**, categorized into **Direct Taxes** (like Income Tax) and **Indirect Taxes** (like Sales Tax/GST). This revenue is then used for **government spending** and **transfer payments** (social safety nets), which stimulate demand in the economy. From a business management perspective, the FBR's policies directly impact a company's bottom line through tax liabilities and influence consumer purchasing power.

Key Takeaways for Revision:

- **SECP Role:** Regulates non-banking corporate sectors, ensuring market transparency and investor protection.
- **SECP Supervision:** Covers registration, licensing, and the administration of professionals like CAs and CMAs.
- **FBR & Fiscal Policy:** Manages the economy via taxation and government expenditure.
- **Tax Categories:** Direct taxes (on income/wealth) and Indirect taxes (on consumption)

Topic 017: The Pakistan Stock Exchange (PSX)

The **Pakistan Stock Exchange (PSX)** serves as the primary hub for **Capital and Money Markets** in the country. It is often described as a "barometer of the economy" because its performance reflects the overall health of the business sector. The PSX facilitates two types of markets: the **Primary Market**, where companies issue new shares to the public

for the first time (Initial Public Offerings), and the **Secondary Market**, where existing securities are traded among investors.

The PSX provides a platform for the **listing of companies** and the **trading of securities**, allowing businesses to raise long-term capital for expansion. For investors, it offers a **Data Portal** and **guidance** to ensure informed decision-making. By attracting **foreign investment**, the PSX helps bring international capital into the local economy. Management must keep a close watch on **Stock Market Indices** (like the KSE-100), as these indices aggregate the performance of top companies and serve as a benchmark for investment performance.

Key Takeaways for Revision:

- **Economic Indicator:** Stock market performance reflects investor confidence and economic health.
 - **Capital Formation:** Facilitates the flow of savings from individuals into corporate investments through primary and secondary markets.
 - **Foreign Investment:** Acts as a gateway for international investors to participate in the local economy.
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Topic 018, 019 & 020: Commercial Banks, DFIs, and NBFCs

The credit landscape is populated by various institutions, each serving a unique purpose. **Commercial Banks** are the most common, governed strictly by the **State Bank of Pakistan**. Their business model revolves around accepting **deposits** (Checking and Saving accounts) and providing **lending** services (Short-term and Long-term loans). They also offer essential financial products like **Certificates of Deposit (CDs)**, mortgages, and unsecured financing, alongside services like fund transfers and insurance. In contrast, **Development Finance Institutions (DFIs)** provide funds for **economic development projects**, particularly in emerging economies. Unlike commercial banks, DFIs often operate on a **non-commercial basis**, taking on **high-risk loans** or equity positions that a traditional bank might avoid. Their goal is not just profit, but national progress through infrastructure and industrial support. They can be **Multilateral** (involving multiple nations) or **National** (focused on a single country). Finally, **Non-Banking Financial Companies (NBFCs)** provide bank-like services but are **not regulated as banks**. They fill gaps in the financial system by offering **commercial loans, factoring, crowdfunding, and venture capital**. Major forms of NBFCs include insurance companies, leasing companies, Modarabas, and microloan providers. They are essential for providing specialized financing like **equity financing** or **joint ventures** that traditional banks may find too risky or outside their core competency.

Key Takeaways for Revision:

- **Commercial Banks:** Core focus on deposit mobilization and standardized lending under SBP rules.
- **DFIs:** Focused on long-term development, often taking high-risk positions for national growth.

- **NBFCs:** Provide specialized financial services (leasing, venture capital, crowdfunding) outside the traditional banking framework.

Topic 021 & 022: Islamic Banking and Foreign Banks

Modern finance also includes specialized banking models. **Islamic Banking** is defined by its strict adherence to **Shariah Compliance**, overseen by dedicated **Shariah Boards**. The fundamental difference is the **prohibition of Riba (interest)**; instead, Islamic banks operate on the principle of **Profit and Loss Sharing**. Common instruments include **Musharika** (partnership), **Modaraba** (profit-sharing investment), **Ijara** (leasing), and **Murabaha** (cost-plus financing).

On the international front, **Foreign Banks** bring global expertise to the host country. Their presence offers several **advantages**, such as increasing the **efficiency of local banks** through competition, providing **sophisticated financial services**, and facilitating access to **international capital markets**. However, there are **disadvantages**, including potential **loss of control** over the domestic financial system and the risk that foreign banks might hinder the **growth of local banks**.

Key Takeaways for Revision:

- **Islamic Banking:** Interest-free finance based on ethical profit-sharing models (Musharika, Modaraba).
- **Foreign Banks Pros:** Innovation, efficiency, and international capital access.
- **Foreign Banks Cons:** Potential for reduced domestic control and competitive pressure on local institutions.

IMPORTANT QUESTIONS FOR EXAM:

Q1. Compare the regulatory roles of the SBP versus the SECP. (5 Marks)

The **SBP (State Bank of Pakistan)** regulates the **banking system and monetary policy**.

Its main roles are:

- controls banks
- issues currency
- controls credit
- manages interest rates
- protects depositors
- lender of last resort

The **SECP (Securities and Exchange Commission of Pakistan)** regulates the **corporate sector and capital markets**.

Its main roles are:

- company registration
- stock market regulation
- investor protection
- insurance sector regulation
- licensing and supervision

Simple Difference

- SBP = banks and money
- SECP = companies and stock market

Q2. Explain the difference between Musharika and Modaraba in Islamic banking. (5 Marks)

Both are Islamic banking methods based on **profit and loss sharing**.

Musharika

In Musharika, **all partners invest money** in the business.
Profit is shared by agreement, and loss is shared according to investment.

Modaraba

In Modaraba, **one party gives money** and the other party **manages the business**.

- investor = provides capital
- manager = runs business
- profit = shared by agreement
- loss = mostly borne by investor

Simple Difference

- Musharika = **both invest**
- Modaraba = **one invests, one manages**

Q3. How does the FBR's fiscal policy impact business financial planning?

The **FBR affects business planning through taxes**.

Businesses must plan for:

- income tax
- sales tax
- GST
- tax filing
- compliance costs

- budgeting for liabilities

Higher taxes increase business costs, while tax relief can improve profits.

It also affects **consumer buying power**, which changes sales planning.

Conclusion

So, FBR policies directly affect **budgeting, profit planning, and financial decisions**.

✦✦ High-Marks Final Line

SBP controls money, SECP controls markets, and FBR controls taxation.

Module 3 (Topics 023–025): The Foundations of Time Value of Money:

The **Time Value of Money (TVM)** is perhaps the most fundamental concept in finance and management. It is based on the premise that a sum of money available now is worth more than the same amount in the future due to its potential earning capacity. In management, this is often viewed through the lens of **opportunity cost**; by having money today, you can invest it to earn interest, whereas receiving it later forces you to defer that consumption or investment.

The two primary pillars of TVM are **Present Value (PV)**—the current value of a future sum—and **Future Value (FV)**—the value a current investment will grow to over a specific period. A critical external factor is **inflation**, defined as the gradual increase in prices for goods and services. While inflation erodes purchasing power, it is important to note that in standard TVM calculations, we do not always incorporate inflation directly, focusing instead on the nominal earning power of interest.

The mechanism by which money grows is **interest**. In an exam, you must distinguish between two types:

1. **Simple Interest:** Interest is calculated solely on the **original principal** amount invested. The formula is $I = P \times r \times t$.
2. **Compound Interest:** Often described as "**interest on interest**," it is calculated on both the principal and the accumulated interest from previous periods. Because the balance increases every time the rate is applied, compound interest **always results in a higher future value** than simple interest over the same duration.

Key Takeaways for Revision:

- **Opportunity Cost:** Choosing money today allows for immediate reinvestment.
- **Inflation:** Decreases the value of money over time but is treated as a separate variable from basic TVM.
- **Simple vs. Compound:** Simple interest stays flat; compound interest grows exponentially because the rate applies to the new balance.

Slide Examples:

- **TVM Choice:** Choosing Rs. 100,000 now (PV) versus Rs. 110,000 after one year (FV) illustrates the growth of money through time.
- **Simple Interest Calculation:** Investing Rs. 100,000 at 10% for 2 years yields Rs. 20,000 interest ($100,000 \times 0.10 \times 2$).

Section 2: Present and Future Value Mechanics (Topics 026–032)

Calculating PV and FV is essential for managers making decisions about capital budgeting or long-term investments. **Compounding** is the process of moving from the present to the future, while **Discounting** is the reverse—determining the current worth of a future payment.

Under **Simple Interest**, the relationship is linear. To find the Future Value, you add the interest to the principal ($FV = PV + I$). Conversely, to find the Present Value, you subtract interest from the future amount ($PV = FV - I$). However, a more robust formula used in management is $PV = FV / (1 + rt)$, where r is the rate and t is time.

Under **Compound Interest**, the growth is non-linear. The FV is calculated by multiplying the PV by the **Future Value Interest Factor (FVIF)**, which is $(1 + r)^n$. When compounding over long periods (e.g., 30 years), the FV becomes significantly higher than simple interest because of the accumulation effect. For discounting, we use the **Discounting Factor** $(1 / (1 + r)^n)$, which helps managers decide if a future payout is worth the current investment cost.

Exam-Critical Relationships:

- **Time Sensitivity:** The higher the number of compounding periods (n), the lower the present value of a future sum.
- **Rate Sensitivity:** As the interest rate (r) increases, the present value decreases.

Slide Examples:

- **Compounding Growth:** Rs. 100,000 at 10% for 2 years under compound interest becomes Rs. 121,000 ($100,000 \times (1.1)^2$), which is Rs. 1,000 more than simple interest.
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- **Discounting Value:** The PV of Rs. 100,000 received in 2 years at 10% is Rs. 82,645.
- **Table Use:** To find the FV of Rs. 100,000 at 3% after 4 years, locate the factor 1.126 in an FVIF table: $100,000 \times 1.126 = 112,600$.

Section 3: Annuities, Perpetuities, and Mixed Flows (Topics 033–041)

In business, cash flows rarely occur as single sums. Managers often deal with **Annuities**—a series of equal payments or receipts at regular intervals, such as insurance premiums or monthly interest income.

- **Ordinary Annuity:** Payments occur at the **end** of each period.
- **Annuity Due:** Payments occur at the **start** of each period (e.g., rent or pension). Because payments happen earlier, an Annuity Due will always have a **higher FV and PV** than an ordinary annuity, as the money has more time to earn interest.

- **Perpetuity:** A series of equal payments that continues **forever**. While the FV of a perpetuity is theoretically infinite, the PV can be calculated ($PV=R/r$) because future installments become so small through discounting that they eventually become negligible.

For scenarios where cash flows are not equal, we encounter **Mixed Cash Flows**. These require calculating the PV or FV of each individual component separately and then summing them up.

Key Takeaways for Revision:

- **FVIFA/PVIFA Tables:** These pre-calculated tables simplify annuity math; you simply multiply the installment (R) by the table factor.
- **Annuity Due Rule:** To adjust an ordinary annuity table factor for an Annuity Due, multiply the FVIFA by $(1+r)$.

Slide Examples:

- **Ordinary Annuity FV:** Saving Rs. 1,000 at the end of every year for 5 years at 10% results in Rs. 6,105 ($1,000 \times 6.105$ FVIFA).
- **Perpetuity PV:** A perpetual payment of Rs. 1,000/year at 10% is worth Rs. 10,000 today.
- **Mixed Flow:** If you receive Rs. 1,000 for 3 years, then Rs. 2,000 for 4 years, you calculate the PV for the first annuity and then the second (discounted further back to year 0) to get the total.

Section 4: Frequency of Compounding and Effective Rates (Topics 042–046)

Standard rates are usually quoted as **Nominal Rates** (per annum). However, if interest is applied more than once a year (**Shorter Period Compounding**), the actual interest earned is higher. This happens because the interest from the first half of the year starts earning its own interest in the second half.

To compare different compounding frequencies, we use the **Effective Rate of Interest (ERI)**. The formula $ERI = \{[1 + (r/m)]^m - 1\} \times 100$ (where m is the number of periods per year) converts a nominal rate into its true annual yield. If compounding happens every second or even faster, it is called **Continuous Compounding**, utilizing the mathematical constant e (2.718).

Management Concept: When a bank offers a 12% rate compounded monthly, the **Nominal Rate** is 12%, but the **Effective Rate** is 12.68%. Managers must use ERI to compare loans from different institutions fairly.

Slide Examples:

- **Shorter Compounding:** Rs. 1,000 at 10% nominal rate becomes Rs. 1,100 if compounded annually, but Rs. 1,102.50 if compounded bi-annually.
- **Continuous FV:** Rs. 1,000 compounded continuously at 10% for 2 years becomes Rs. 1,221.38.
- **Equivalent Rate:** If you want a quarterly rate that results in the same annual growth as a 12% nominal rate, you use the formula $((1+r)^{1/n} - 1)$ to find the equivalent rate of 2.87%.

Section 5: Strategic Applications: Amortization and Sinking Funds (Topics 047–051)

Loan Amortization is the process of paying off a debt in equal installments. Each payment covers both **interest** (calculated on the remaining balance) and a portion of the **principal**. In the early years of a loan, most of the installment goes toward interest; as the balance drops, more goes toward the principal.

A **Sinking Fund** is the opposite of a loan. It is a savings plan where a business sets aside equal amounts regularly to meet a future obligation, such as replacing an asset or retiring bonds. To calculate the required deposit, you use the **Future Value of an Annuity** formula, solving for the installment (R).

Key Takeaways for Revision:

- **Amortization Schedule:** Tracks the declining balance and ensures the loan hits zero at the end of the term.
- **Sinking Fund Advantage:** Setting up a sinking fund as an **Annuity Due** (deposits at the start of the year) is more efficient because it earns more interest, meaning the company has to deposit less money to hit its goal.

Slide Examples:

- **Amortization:** Borrowing Rs. 100,000 at 10% for 5 years requires an annual installment of Rs. 26,379.75.
- **Sinking Fund:** To save Rs. 100,000 in 5 years at 10% (Ordinary Annuity), you must deposit Rs. 16,379.75 annually.
- **Sinking Fund (Annuity Due):** For the same goal, depositing at the start of the year reduces the required installment to Rs. 14,890.68

IMPORTANT DEFINITIONS (TIME VALUE OF MONEY)

1. Time Value of Money (TVM)

Time Value of Money means that **money available today is worth more than the same amount in the future** because it can earn interest and grow over time.

2. Interest

Interest is the **extra amount earned or paid for using money over time**.

3. Simple Interest

Simple interest is the interest calculated **only on the original principal amount**, not on accumulated interest.

4. Compound Interest

Compound interest is the interest calculated on **both principal and previously earned interest**.

5. Present Value (PV)

Present value is the **current worth of a future amount of money discounted at a given interest rate**.

6. Future Value (FV)

Future value is the **value of a present amount of money at a specific time in the future after earning interest**.

7. Discounting

Discounting is the process of **finding the present value of future cash flows**.

8. Compounding

Compounding is the process of **earning interest on both principal and accumulated interest over time**.

9. Annuity

An annuity is a **series of equal payments made at equal time intervals**.

10. Ordinary Annuity

An ordinary annuity is a series of equal payments made **at the end of each period**.

11. Annuity Due

An annuity due is a series of equal payments made **at the beginning of each period**.

12. Perpetuity

Perpetuity is an annuity that **continues forever with no end date**.

13. Mixed Cash Flows

Mixed cash flows are **unequal and irregular cash inflows and outflows occurring at different times**.

14. Shorter Period Compounding

It is the process where interest is **compounded more than once in a year (semi-annually, quarterly, monthly)**.

15. Continuous Compounding

Continuous compounding is when interest is **compounded infinitely many times in a year**.

16. Effective Interest Rate (ERI)

Effective interest rate is the **actual annual rate earned or paid after considering compounding effects**.

17. Nominal Interest Rate

Nominal interest rate is the **stated annual interest rate without considering compounding within the year.**

18. Loan Amortization

Loan amortization is the process of **repaying a loan through regular installments that include both interest and principal.**

19. Amortization Schedule

An amortization schedule is a **table showing breakdown of each loan payment into interest, principal, and remaining balance.**

20. Sinking Fund

A sinking fund is a **fund created by saving a fixed amount regularly to repay a future liability or replace an asset.**

21. Future Value Interest Factor (FVIF)

FVIF is a **multiplier used to calculate future value of a present amount using compound interest.**

22. Present Value Interest Factor (PVIF)

PVIF is a **multiplier used to calculate present value of a future amount.**

23. Annuity Factor

An annuity factor is a **formula used to calculate present or future value of a series of equal payments.**

FINAL EXAM TIP

If you memorize only this structure:

- PV = present worth
- FV = future worth
- Annuity = equal payments
- Perpetuity = forever payments
- Compounding = growth
- Discounting = reverse growth

FINAL QUICK MEMORY

- ✓ FV → multiply → $(1+r)^n$
 - ✓ PV → divide → $(1+r)^n$
 - ✓ Annuity → divide by r
 - ✓ Perpetuity → R / r
-

Q1. How do taxes impact business management and fiscal policy?

Taxes affect business management by changing the **costs, profits, and financial planning** of a company.

Businesses must include taxes in:

- budgeting
- pricing decisions
- profit planning
- cash flow management
- investment decisions

In **fiscal policy**, taxes help the government collect revenue for spending on development and welfare.

Higher taxes may reduce business profit, while lower taxes can encourage investment.

Conclusion

So, taxes directly affect **business decisions and government economic policy**.

Q2. How does a loan amortization schedule track declining balance?

A loan amortization schedule shows how a loan is paid back in **equal installments over time**.

Each payment has:

- **interest portion**
- **principal portion**

At the start:

- more payment goes to **interest**
- less goes to principal

As time passes:

- loan balance decreases
- interest amount becomes smaller
- more payment goes to principal

This process continues until the **loan balance becomes zero**.

Conclusion

So, the schedule tracks the **remaining balance after every installment**.

High-Marks Final Line

Amortization helps businesses manage debt repayment in a planned way.



Module 4 Topic 053–055:

Foundations of Securities and the Concept of Valuation:

In financial management, a **security** is defined as a financial instrument that is interchangeable, tradable, and possesses monetary value. Essentially, these "pieces of paper" represent a legal claim of an owner on the assets and future cash flows of the issuing company. Securities are categorized by their **term** (duration): **Short-term securities** (like T-Bills, CDs, and Repos) have a maturity of less than one year and are exclusively debt-based, while **long-term securities** (like Stocks and Bonds) have a maturity exceeding one year. Management focuses on long-term securities because they form the core of a firm's **capital structure**, consisting of either **Equity Capital** (ownership) or **Debt Capital** (borrowed funds).

Valuation is the analytical process of determining the current worth of an asset or a security. As Oscar Wilde noted, it is vital to know the "value" rather than just the "price". For a corporation, valuation is essential during **Book Building** to set a fair issue price for new securities. For an investor, the goal is to identify the **Intrinsic Value**—the "true" or "real" value based on internal factors—and compare it to the **Market Price**. This comparison prevents investors from buying "overpriced" (over-sold) or "underpriced" (over-bought) securities, thereby optimizing the risk-return profile of their portfolio. Understanding the different "values" of a security is a common exam requirement. These include:

- **Face/Par Value:** The value written on the certificate, used to calculate interest or dividends.
- **Market Value:** The current price at which the security trades in the stock exchange.
- **Book Value:** The value of a share according to accounting records ($\text{Total Equity} \div \text{Number of Shares}$).
- **Liquidation vs. Going Concern Value:** Liquidation value is what remains if a company is wound up (usually much lower), while **Going Concern Value** assumes the business will continue indefinitely.
- **Break-up Value:** The value realized if a large company is split into smaller, potentially more profitable segments.

Key Takeaways for Revision:

- **Security:** A tradable financial instrument representing a claim on assets.
- **Long-term:** Maturity > 1 year; includes Equity (Stocks) and Debt (Bonds).
- **Intrinsic Value:** The core objective of valuation; used to make "Buy/Hold/Sell" decisions.
- **Going Concern:** The assumption that the business will not discontinue in the foreseeable future.

Topic 057–060: The Valuation of Bonds (Fixed Income Securities)

Bonds are **debt securities** that provide investors with **Fixed Income** in the form of periodic interest payments, known as **coupons**. To value a bond, a manager must calculate the **Present Value (PV)** of all future cash inflows the bond will generate, discounted at a **Required Rate of Return** (the investor's opportunity cost). There are three primary types of bonds you must be able to value for the exam:

1. **Perpetual Bonds:** These bonds have no maturity date and pay interest forever. The valuation formula is the simplest, based on the perpetuity model: $V = \frac{I}{r}$, where I is the annual interest and r is the required rate. If the intrinsic value (V) is higher than the market price, the bond is a good buy.
2. **Zero-Coupon Bonds:** These bonds do not pay periodic interest; instead, they are issued at a discount and pay a single **Maturity Value (MV)** at the end of their term. To find their value, we use the basic PV formula: $V = \frac{MV}{(1+r)^n}$. Since there are no interim cash flows, the value of these bonds is highly sensitive to changes in interest rates and the time to maturity (n).
3. **Coupon Bonds (Definite Maturity):** This is the most common bond type. Its value is the sum of two components: the **PV of periodic interest payments** (calculated as an annuity) and the **PV of the maturity value** (calculated as a single future amount).

Practical Application: If a bond has a 10% coupon rate but the market's required rate is 13%, the bond will sell at a **discount** (less than face value) because its fixed payments are less attractive than current market opportunities.

Key Takeaways for Revision:

- **Bond Components:** Face Value, Maturity Period, Coupon Rate, and Required Rate.
- **Perpetual Formula:** $V = \frac{I}{r}$.
- **Zero-Coupon Formula:** $V = \frac{MV}{(1+r)^n}$.
- **Coupon Bond Valuation:** Discount both the annual interest and the final principal.

Topic 061–064: Preferred Stock Valuation

Preferred Stock is a "hybrid" security, possessing characteristics of both bonds and common stock. Like bonds, they usually offer a **fixed dividend rate** and have a higher claim on assets than common stockholders during liquidation. However, like common stock, they represent equity and usually do not have a fixed maturity date.

For your exam, pay close attention to the specific **features** that can affect value:

- **Cumulative:** If the company misses a dividend, it accumulates and must be paid in the future before common shareholders receive anything.
- **Participative:** Allows shareholders to receive extra dividends if company profits exceed a certain level.
- **Convertible:** Gives the holder the option to exchange preferred shares for a set number of common shares, providing potential for capital gains.
- **Redeemable/Callable:** The issuer has the right to "call" or buy back the stock at a specific price after a certain time.

Valuation Model: Since preferred stock dividends are typically constant and paid indefinitely (non-redeemable), we use the **Perpetuity Formula:** $V = \frac{D}{r}$, where D is the constant dividend and r is the required rate. If your required rate (r) is higher than the dividend rate, the intrinsic value will naturally be lower than the par value.

Key Takeaways for Revision:

- **Priority:** Preferred stockholders are paid dividends before common stockholders.
 - **No Voting Rights:** Unlike common stock, preferred shares usually do not grant voting power.
 - **Valuation:** Use the D/r formula, assuming dividends are constant and infinite.
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Topic 065–072: Common Stock Valuation Models

The valuation of **Common Stock** is more complex because dividends are not fixed and depend on company performance. Common shareholders are the "true owners" who bear the highest risk but also enjoy voting rights and unlimited growth potential. We use the **Dividend Discount Model (DDM)**, which assumes the value of a stock is the PV of all future dividends.

There are three scenarios for common stock growth that you must master:

1. **No Growth Model:** Assumes the company pays out all earnings as dividends and the dividend stays constant forever ($g=0$). The formula is identical to preferred stock: $V = \frac{D}{r}$.
2. **Constant Growth Model (Gordon Growth Model):** Assumes dividends grow at a constant rate (g) forever. The formula is: $V = \frac{D_1}{r-g}$, where D_1 is the dividend expected next year [$D_0 \times (1+g)$]. A critical assumption here is that $r > g$; if the growth rate were higher than the required rate, the formula would break down.
3. **Multi-Stage Growth Models (Two-Stage & Three-Stage):** These models reflect real-world companies that might experience high growth initially before stabilizing.
 - **Stage I:** Calculate the PV of each individual dividend during the high-growth period.
 - **Final Stage:** Use the Constant Growth Model to find the value of all dividends from that point to infinity (Terminal Value), and then discount that value back to the present.
 - **Total Value:** Sum the PVs of all stages.

Key Takeaways for Revision:

- **Gordon Growth Formula:** $V = \frac{D_1}{r-g}$.
- **Next Year's Dividend (D_1):** Always use the expected dividend, not the last dividend (D_0).
- **Decision Rule:** Buy if **Intrinsic Value $\geq$ Market Price**

Module 5: Financial Analysis Tools

Unit 1: The Foundation of Financial Reporting

To conduct a meaningful financial analysis, one must first understand the primary "data sources": the financial statements. These documents provide a structured representation of a company's economic health. The **Statement of Financial Position (Balance Sheet)** provides a snapshot of a company's resources (Assets) and the claims against those resources (Liabilities and Owner's Equity) at a specific point in time. Assets are categorized by their liquidity: **Current Assets** (expected to be converted to cash within a year) and **Non-Current Assets** (long-term investments like Operating Fixed Assets). Liabilities follow a similar structure, divided into **Current Liabilities** (short-term obligations) and **Long-term Liabilities**. **Owner's Equity** represents the residual interest of the shareholders, including issued capital, reserves, and **Retained Earnings**—which are profits reinvested back into the business rather than paid out as dividends. Complementing this is the **Statement of Comprehensive Income (P&L Account)**, which measures financial performance over a period. From a management perspective, the focus is on various levels of profit: **Gross Profit** (Revenue minus Cost of Sales), **EBIT** (Earnings Before Interest and Tax), and **EAT** (Earnings After Tax). A critical management concept here is **EBITDA** (Earnings Before Interest, Tax, Depreciation, and Amortization), which serves as a proxy for operating cash flow and allows for comparison between firms with different capital structures and tax environments. Finally, the **Statement of Cashflows** is essential because profit does not equal cash. It tracks cash movements across three activities: **Operating** (core business), **Investing** (buying/selling assets), and **Financing** (debt and equity changes).

Key Takeaways for Revision

- **Balance Sheet:** Assets = Liabilities + Equity (snapshot of resources).
- **Retained Earnings:** Cumulative profits kept in the business for growth.
- **EBITDA:** A vital "clean" measure of operating performance excluding non-cash and non-operating items.
- **Cash Flow Statement:** Essential to distinguish between accounting profit and actual cash availability.

Unit 2: Framework for Financial Analysis & Benchmarking

Financial analysis is the process of evaluating a firm's financial statements to make better economic decisions. It is categorized into **Absolute Analysis** (looking at raw figures) and **Relative Analysis** (using ratios and indexes). In management, **Relative Analysis** is superior because it provides context; a profit of \$1 million is impressive for a small shop but poor for a global corporation. Common methods include **Ratio Analysis**, **Trend Analysis** (comparing performance over time), and **Common Size Analysis** (expressing items as a percentage of a base figure).

A core component of professional analysis is **Benchmarking**. Ratios in isolation are meaningless; they must be compared against **Past Performance**, **Industry Averages**, or

Target Ratios set by management. Benchmarking helps stakeholders like **Lenders** (evaluating risk), **Shareholders** (evaluating returns), and **Management** (evaluating efficiency). For example, a company might use benchmarking during **Mergers** or when **Going Public** to ensure their valuation is competitive.

Key Takeaways for Revision

- **Absolute vs. Relative:** Ratios provide the necessary context that raw numbers lack.
 - **Primary Users:** Management, Creditors, and Investors all use these tools for different risk/reward assessments.
 - **Benchmarks:** Always compare against the industry average or the company's own historical trends to identify deviations.
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Unit 3: Liquidity and Activity Analysis

Liquidity refers to a company's ability to meet its short-term obligations as they fall due. Failure to maintain liquidity leads to **Technical Insolvency**, even if the company is profitable on paper. The primary measures are the **Current Ratio** (Current Assets / Current Liabilities) and the **Quick Ratio** (excluding inventory and prepayments, which are harder to liquidate). A general standard for the Current Ratio is 1.5 to 2.0, while the Quick Ratio should ideally be 1.0.

Beyond simple ratios, management focuses on the **Cash Conversion Cycle (CCC)**, which measures the time (in days) it takes to convert resource inputs into cash flows. The formula is **RTD (Receivables Days) + ITD (Inventory Days) - PTD (Payables Days)**. A shorter CCC is generally better, as it indicates the business is not "blocking" cash in the operating cycle for too long. **Activity or Turnover Ratios**—such as Inventory Turnover and Receivables Turnover—measure how efficiently the firm uses its assets. For example, a low **Inventory Turnover** suggests overstocking or obsolete goods, which increases holding costs.

Key Takeaways for Revision

- **Liquidity Standards:** Current Ratio (1.5–2.0) and Quick Ratio (1.0) are the typical benchmarks.
 - **Cash Conversion Cycle:** Represents the number of days cash is tied up in operations; lower is usually more efficient.
 - **Stretching Payables:** Delaying payments to creditors to save cash, but at the risk of losing credibility or discounts.
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Unit 4: Leverage, Coverage, and Profitability

Financial Leverage (or Gearing) involves using fixed-cost debt to increase the potential return on equity (ROE). While debt is often cheaper than equity, it introduces **Fixed**

Financial Costs and increases risk. A classic management example shows that if a company's **Return on Investment (ROI)** (e.g., 20%) is higher than its **Cost of Debt** (e.g., 10%), leverage will boost the **ROE** (e.g., to 25%). Common ratios include **Debt to Equity** and **Debt to Total Assets**.

To manage the risk of high leverage, we use **Coverage Ratios**. The **Interest Coverage Ratio (ICR)** (EBIT / Interest) measures how easily a company can pay interest on outstanding debt. The **Debt Service Coverage Ratio (DSCR)** is more rigorous, as it includes both interest and principal repayments, adjusted for taxes. Finally, **Profitability Analysis** looks at margins (e.g., **Gross Profit Margin**, **Net Profit Margin**) and returns (e.g., **ROA**, **ROE**). Management uses these to assess whether the revenue generated is effectively being converted into profit after all costs and taxes are accounted for.

Key Takeaways for Revision

- **Leverage Principle:** Debt magnifies returns if $ROI > \text{Cost of Debt}$, but it also increases the risk of bankruptcy.
- **Coverage:** ICR and DSCR are vital for lenders to assess a firm's ability to "service" its debt.
- **Profitability vs. Profit:** Profit is an absolute amount; profitability is a relative measure of efficiency (Margins and Returns).

Unit 5: Market Ratios and Comparative Analysis

For investors, **Market Ratios** are the ultimate yardstick. **Earnings Per Share (EPS)** ($PAT / \text{Number of Shares}$) and **Dividend Per Share (DPS)** tell an investor exactly what they are "earning" per unit of ownership. The **Price/Earnings (P/E) Ratio** is a critical valuation tool, showing how much investors are willing to pay for each dollar of earnings; a high P/E often indicates expected future growth. Other metrics like **Dividend Yield** and **Earning Yield** allow investors to compare the "return" on a stock to other investment opportunities like bonds.

To perform high-level comparisons, analysts use **Index Analysis** and **Common Size Analysis**. **Index Analysis** sets a "Base Year" to 100 and expresses subsequent years as a percentage of that base, making it easy to spot growth trends (e.g., Sales growing by 12.73% while COGS grows by 28.72% indicates a margin squeeze). **Common Size Analysis** (Vertical Analysis) expresses every line item as a percentage of a base—usually **Sales** for the P&L and **Total Assets** for the Balance Sheet. This allows for the comparison of companies of different sizes. Lastly, **Differential Analysis** is used for decision-making (e.g., "Drop a product line?"), focusing only on **Relevant Costs** (future cash flows and opportunity costs) while ignoring **Sunk Costs** (past expenditures that cannot be recovered).

Key Takeaways for Revision

- **P/E Ratio:** A primary indicator of market confidence and growth expectations.
- **Common Size Analysis:** Vital for comparing a small firm to a large firm by neutralizing the "size" factor.

- **Differential Analysis:** Decision-making should only focus on "Relevant Costs" that will change based on the decision.

IMPORTANT QUESTIONS FOR EXAM

1. How do we calculate the Cash Conversion Cycle?

The **Cash Conversion Cycle (CCC)** is calculated using the formula: $CCC = RTD + ITD - PTD$. This metric represents the number of days it takes for a company's economic activity to be converted into cash. Specifically, **RTD** is Receivables Turnover in days, **ITD** is Inventory Turnover in days, and **PTD** is Payables Turnover in days.

2. What is the difference between Horizontal and Vertical Analysis?

The difference between Horizontal and Vertical Analysis is:

- **Horizontal Analysis** (also known as Index Analysis) evaluates financial data over time by comparing current values to a **base year** (converted to 100) to identify growth or decline trends.
- **Vertical Analysis** (Common Size Analysis) evaluates a single period by expressing every item as a **percentage of a base value**, such as **Total Assets** for the Balance Sheet or **Sales** for the P&L Account.

3. Why are Sunk Costs ignored in Differential Analysis?

Sunk costs are ignored in **Differential Analysis** because they are classified as **non-relevant costs**. Differential Analysis focuses exclusively on **relevant costs**—such as future cash flows and opportunity costs—that will change based on the decision being made. Since sunk costs are past expenditures that cannot be recovered or altered by future actions, they do not impact the comparison between different investment or production options

Module 6: Cash Flow Analysis

Unit 1: The Philosophy of Cash Flow vs. Profitability

The foundation of financial health lies in understanding that **profit is not cash**. While the Statement of Comprehensive Income measures performance based on the accrual principle (recognizing revenue when earned and expenses when incurred), the **Statement of Cash Flows** tracks the actual movement of currency into and out of the business.

Cash Inflows primarily stem from revenues, borrowings, and equity issuances, while **Cash Outflows** consist of expenses, debt repayments, and dividends.

From a management perspective, a company can report a high net profit but still face bankruptcy if that profit is "trapped" in accounts receivable or unsold inventory.

Managers must analyze **Sources of Cash** (e.g., collection from receivables, sale of assets, or issuing shares) and the **Application of Cash** (e.g., purchasing fixed assets, paying down liabilities, or treasury stock buybacks) to ensure the firm maintains **Liquidity**. A firm that generates cash from its core operations is generally seen as more sustainable than one that relies on selling assets or constant borrowing to stay afloat.

Key Takeaways for Revision

- **Profit vs. Cash:** Profit is an accounting construct; cash flow is the literal movement of money.
- **Solvency Risk:** High profits do not guarantee the ability to pay bills if cash is tied up in non-cash assets.
- **Sources & Applications:** Management must balance where money comes from (e.g., bonds, equity) with where it goes (e.g., capital expenditure, dividends).

Unit 2: Operating Activities and the Indirect Method

Cash Flow from Operating Activities is arguably the most important section of the statement because it reveals the cash-generating power of the firm's core business. The **Indirect Method** begins with **Net Income** and "cleanses" it of non-cash items and working capital distortions. The first step is adding back **Non-Cash Expenses**—such as **Depreciation, Amortization, and Provisions for Bad Debts**—because these reduce reported profit but do not involve an actual cash outflow.

The second step involves analyzing **Changes in Current Assets and Liabilities**. In an exam, remember this logic: An **increase in a Current Asset** (like Accounts Receivable or Inventory) is a **Cash Outflow** because it represents cash "blocked" in the business cycle. Conversely, a **decrease in a Current Asset** is a **Cash Inflow** as that asset is converted back to cash. For **Current Liabilities**, the relationship is direct: an **increase in Accounts Payable** is a **Cash Inflow** (as you are "borrowing" from suppliers by delaying payment), while a **decrease in Liabilities** is a **Cash Outflow** (as you are using cash to settle debts). For example, if a firm has a Net Income of Rs. 100,000 and Depreciation of

Rs. 20,000, but its Accounts Receivable increased by Rs. 18,000, the net cash inflow from operations would be adjusted accordingly.

Key Takeaways for Revision

- **The Add-Back Rule:** Always add back non-cash charges like depreciation and amortization to net income.
 - **Asset/Liability Logic:** Assets are inverse to cash (Increase = Outflow); Liabilities are direct to cash (Increase = Inflow).
 - **Inventory & Receivables:** Managing these is key to improving operating cash flow; lower inventory levels free up cash.
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Unit 3: Investing and Financing Activities

Beyond daily operations, management must handle long-term growth and capital structure. **Investing Activities** focus on changes in **Non-Current Assets**. This includes the purchase or sale of **Fixed Assets** (land, buildings, equipment), **Intangible Assets**, and **Long-term Investments** (like buying bonds of another company). For example, if a company spends Rs. 5,000 on land and Rs. 4,000 on a building, these are recorded as cash outflows. A critical exam point is calculating the cash flow from asset changes by considering **Accumulated Depreciation**; the actual cash spent is the change in net assets plus the depreciation for the period.

Financing Activities deal with how the firm is funded. **Inflows** in this category come from issuing **Common or Preferred Shares**, taking out **Loans**, or issuing **Bonds**.

Outflows occur when the firm pays **Dividends**, repays the principal on loans, or buys back its own shares (**Treasury Stock**). It is vital to note that while Net Income is the starting point for operating activities, it is not part of financing activities because its components have already been treated in the operating section. This section reflects the firm's strategy regarding debt vs. equity and its commitment to returning value to shareholders.

Key Takeaways for Revision

- **Investing Scope:** Focuses on the "Left side" of the Balance Sheet—Long-term Assets and Investments.
 - **Financing Scope:** Focuses on the "Right side" of the Balance Sheet—Long-term Debt and Equity.
 - **Dividend Treatment:** Dividends paid are a financing outflow, representing a cost of equity capital.
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Unit 4: The Direct Method – A Granular Approach

While the Indirect Method is more common in practice, the **Direct Method** provides a clearer view of specific cash receipts and payments. Instead of starting with profit, it lists actual cash exchanged with stakeholders. To calculate these figures for an exam, you must adjust P&L figures with their related Balance Sheet accounts. For instance, **Cash Receipts from Customers** is calculated as **Net Sales + Opening Receivables – Ending Receivables**. This formula accounts for sales made on credit that haven't been collected yet.

Similarly, payments to external parties follow a specific logic. **Cash Paid to Suppliers** involves adjusting **Purchases** for changes in **Inventory** and **Accounts Payable**.

Payments to employees are calculated by taking **Salaries Expense** and adjusting for changes in **Salaries Payable**. Other calculations include **Cash for IT Payment** (Income Tax Expense adjusted for Tax Payable) and **Interest Payments**. This method is highly transparent, as it shows exactly how much cash was paid for rent, insurance, taxes, and interest, making it a powerful tool for short-term cash budgeting and creditor analysis.

Key Takeaways for Revision

- **Granular Focus:** Shows actual cash "to" and "from" specific groups (Customers, Suppliers, Employees).
- **Core Formulas:**
 - From Customers: $\text{Sales} + \text{Opening A/R} - \text{Closing A/R}$.
 - To Suppliers: $\text{Purchases} + \text{End Stock} - \text{Open Stock} + \text{Opening A/P} - \text{Closing A/P}$.
- **Decision Tool:** Preferred by some analysts because it removes the "noise" of accounting adjustments found in the indirect method.

IMPORTANT QUESTIONS FOR EXAM

Q1. How do you calculate cash flow from asset changes?

Cash flow from asset changes is calculated in **investing activities**.

Formula

$\text{Cash Outflow} = (\text{Ending Asset} - \text{Beginning Asset}) + \text{Depreciation}$

Explanation

- Increase in assets means **cash outflow** (purchase of assets)
- Depreciation is added back because it is **non-cash expense**

Conclusion

This shows the **actual cash spent on long-term assets**.

Q2. Why are Retained Earnings excluded from Financing Activities?

Retained Earnings are excluded to **avoid double counting**.

- Retained Earnings come from **Net Income**
- Net Income is already included in **Operating Activities**
- Financing activities only include **external funds** (shares, loans)

Conclusion

So, they are excluded because they are **internal profit already counted earlier**.

Q3. How do I calculate Cash Receipts from Customers?

Formula

Cash Receipts = Net Sales + Opening Accounts Receivable – Ending Accounts Receivable

Explanation

- Add opening receivables (money received from past sales)
- Subtract ending receivables (sales not yet collected)

Conclusion

This shows the **actual cash received from customers**.

Q4. What is the Cash Conversion Cycle formula?

The Cash Conversion Cycle (CCC) measures how fast a company converts its investment into cash.

Formula

CCC = Inventory Days + Receivable Days – Payable Days

Explanation

- Inventory Days → time to sell goods
- Receivable Days → time to collect cash
- Payable Days → time to pay suppliers

Conclusion

A shorter CCC means **better cash management**.

Q5. Explain the difference between Horizontal and Vertical Analysis.

Horizontal Analysis

- compares financial data over time
- shows growth or decline
- example: comparing sales of 2023 and 2024

Vertical Analysis

- shows each item as a percentage of total
- used in one period
- example: expenses as % of sales

Conclusion

- Horizontal = **trend over time**
 - Vertical = **percentage within one year**
-

Q6. Why are Sunk Costs ignored in Differential Analysis?

Sunk costs are **past costs that cannot be changed**.

- already spent money
- not relevant for future decisions
- do not affect new choices

Example

Money spent on old machinery cannot be recovered.

Conclusion

So, sunk costs are ignored because they **do not impact future decisions**.

✦✦ High-Marks Final Line

Good financial decisions focus only on relevant future costs and cash flows.

Module 7: Risk & Return



Unit 1: The Fundamental Philosophy of Risk and Return

The relationship between risk and return is the cornerstone of modern financial theory. In the simplest terms, a **Return** is the additional amount generated on an investment, which typically manifests as **Dividends** from stocks or **Interest** from debt instruments. However, management must distinguish between the "expected" return and the "realized" return. The primary challenge in finance is **Uncertainty**, which represents the possibility that the actual return will differ from the expected outcome. To quantify this, analysts use the **Expected Return (Mean)** to represent the central tendency of potential outcomes and the **Standard Deviation** to measure the dispersion or "volatility" around that mean.

A critical management concept is that **Risk and Return are Proportional**. This means that an investor cannot expect a higher return without accepting a higher level of uncertainty. In a competitive market, assets are priced based on this trade-off. For example, a government bond is low-risk and thus offers a low interest rate, whereas a startup's equity carries high uncertainty but offers the potential for massive dividends. Exam-focused writing should emphasize that "risk" in finance is not just the "chance of loss," but the **volatility** of returns—the degree to which they fluctuate.

Key Takeaways for Revision

- **Return Components:** Dividends (equity) and Interest (debt) are the primary rewards for capital.
- **Risk Definition:** Uncertainty and the deviation from the expected mean.
- **The Proportionality Rule:** Higher potential returns strictly require higher risk tolerance.

Unit 2: Systematic vs. Unsystematic Risk and the Power of Diversification

To manage a portfolio effectively, a manager must categorize risk into two distinct types: Systematic and Unsystematic. **Systematic Risk**, also known as **Market Risk**, **Undiversifiable Risk**, or **Unavoidable Risk**, is the volatility caused by factors that affect the entire economy. These include interest rate changes, inflation, recessions, or political instability. Because these factors impact all companies simultaneously, this risk

cannot be eliminated through diversification. In the exam, it is crucial to note that **Beta** (β) is the primary index used to measure this systematic risk. In contrast, **Unsystematic Risk** is **Avoidable**, **Diversifiable**, and **Controllable**. This risk is unique to a specific company or industry, such as a labor strike, a faulty product launch, or a change in management. By holding a diversified portfolio of 15–20 stocks across different sectors, an investor can "cancel out" these company-specific events. Therefore, the market does not reward investors for taking on unsystematic risk; it only rewards them for the systematic risk they cannot avoid. A "clean" management strategy focuses on maximizing return for a given level of undiversifiable market volatility.

Key Takeaways for Revision

- **Systematic Risk:** External, market-wide, and undiversifiable (e.g., inflation).
 - **Unsystematic Risk:** Internal, company-specific, and can be eliminated through diversification.
 - **Volatility:** The core measure of risk that managers track via Standard Deviation and Beta.
-

Unit 3: Statistical Tools for Risk Analysis: Mean, SD, and CV

Measuring the risk of an individual stock requires precise statistical tools. The **Mean** (Expected Return) is calculated by multiplying each potential return by its probability. **Variance** measures the squared deviations from that mean, and the **Standard Deviation (SD)** is the square root of variance, providing a measure of risk in the same units as the return. A higher SD indicates a "riskier" stock because its returns are more spread out. However, comparing two stocks based on SD alone can be misleading if their expected returns are different. For this, managers use the **Coefficient of Variation (CV)**, which measures **relative dispersion** or risk per unit of expected return. The formula is **CV = Standard Deviation / Mean**. For example, if Company A has a 10% return with 8% risk (CV = 80%) and Company B has a 20% return with 12% risk (CV = 60%), Company B is actually the better "risk-adjusted" choice because it provides more return for every "unit" of risk taken. In an exam, always use CV to compare the efficiency of different investments.

Key Takeaways for Revision

- **Standard Deviation Formula:** The square root of the sum of squared deviations times their probability.
 - **Coefficient of Variation (CV):** The ultimate tool for comparing stocks with different return levels; it measures "risk per unit of return".
 - **Relative Risk:** A lower CV indicates a more efficient investment.
-

Unit 4: Portfolio Risk and the Capital Asset Pricing Model (CAPM)

When assets are combined, **Portfolio Risk** is not just a simple average of individual risks; it depends on the **Correlation** (ρ) and **Covariance** between assets. **Correlation** measures the mutual relationship (from -1 to +1), while **Covariance** measures the direction of the relationship. If two stocks move in opposite directions (negative correlation), they significantly reduce the overall portfolio risk. The complex equations for two-stock and three-stock portfolios account for the "inter-connectedness" of these assets.

The **Capital Asset Pricing Model (CAPM)** is the most widely used model to determine the required rate of return. It states that the return on a stock equals the **Risk-Free Rate** (R_f) plus a **Risk Premium**. The Risk Premium is the **Market Risk Premium** ($R_m - R_f$) multiplied by the stock's **Beta** (β). **Beta** represents the stock's sensitivity to the market: a Beta of 1.0 moves exactly with the market, a Beta > 1.0 is "aggressive" (more volatile), and a Beta < 1.0 is "defensive" (less volatile). The **Security Market Line (SML)** graphically represents this model, showing the required return for any given level of Beta.

Key Takeaways for Revision

- **CAPM Formula:** $Return = R_f + (R_m - R_f)\beta$.
- **Beta (β):** The slope of the characteristic line; it measures sensitivity to systematic market movements.
- **Risk-Free Rate (R_f):** The return on an investment with zero risk (e.g., T-Bills).

Unit 5: Advanced Factor Models and Market Efficiency

Modern finance has evolved beyond the single-factor CAPM. The **Fama and French Three-Factor Model** adds **Size Risk** (Small Cap minus Big Cap) and **Value Risk** (High minus Low Book-to-Market value) to the equation, recognizing that small-value stocks often outperform the broader market. The **Carhart Four-Factor Model** further adds **Momentum**, which is the tendency for "winning" stocks to continue rising and "losers" to continue declining. These multiple-factor models provide a more accurate "risk-adjusted" expected return for complex portfolios like hedge funds.

Finally, the **Efficient Market Hypothesis (EMH)** suggests that stock prices reflect all available information, making it impossible to consistently "beat the market". The EMH has three levels: **Weak Form** (historical prices), **Semi-Strong Form** (public information), and **Strong Form** (all public and private information). Complementing this is **Arbitrage Pricing Theory (APT)**, which suggests that if an asset is mispriced, the process of **Arbitrage** (buying low in one market and selling high in another) will quickly bring the price back to equilibrium.

Key Takeaways for Revision

- **Fama-French:** Focuses on Market, Size, and Value factors.
- **Momentum (UMD):** The "Up minus Down" factor based on recent price trends.

- **EMH Levels:** Weak (past data), Semi-Strong (public data), and Strong (all data).
- **Arbitrage:** The mechanism that ensures market efficiency by correcting price discrepancies.

IMPORTANT QUESTIONS FOR EXAM

Q1. What is the difference between Systematic and Unsystematic Risk?

Systematic Risk

- affects the **whole market/economy**
- examples: inflation, interest rate, recession
- **cannot be reduced** by diversification

Unsystematic Risk

- affects **one company or industry only**
- examples: strike, bad management, product failure
- **can be reduced** by diversification

Simple Difference

- Systematic = **market risk (unavoidable)**
 - Unsystematic = **company risk (avoidable)**
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Q2. Explain the three forms of Efficient Market Hypothesis (EMH).

The **Efficient Market Hypothesis (EMH)** says stock prices reflect information.

Three Forms

1. **Weak Form**
Prices reflect **past data** (old prices)
2. **Semi-Strong Form**
Prices reflect **all public information** (news, reports)
3. **Strong Form**
Prices reflect **all information** (public + private)

Conclusion

It means it is very difficult to **consistently beat the market**.

Q3. How do you calculate the Coefficient of Variation (CV)?

Formula

$CV = \text{Standard Deviation} / \text{Mean (Expected Return)}$

Explanation

- shows **risk per unit of return**
- helps compare different investments

Example

- lower CV = better investment
- higher CV = more risk

Conclusion

CV is used to choose the **most efficient investment with less risk.**

✦✦ High-Marks Line

Investors prefer low risk with high return, so lower CV is always better

NOTE: These handouts are detail explanation of ppt slides of mgt201 provided by vu .
these are made with help of AI for your ease and better understanding of each and every topic.

Best of luck!

Regards = Tasha khanam